

National Coverage Determination (NCD)

Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)

20.16

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Tracking Information

Publication Number

100-3

Manual Section Number

20.16

Manual Section Title

Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)

Version Number

2

Effective Date of this Version

01/23/2004

Ending Effective Date of this Version

11/24/2006

Implementation Date

02/23/2004

Description Information

Benefit Category

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

Thoracic electrical bioimpedance (TEB) devices, a form of plethysmography, monitor cardiac output by noninvasively measuring hemodynamic parameters, including: stroke volume, systemic vascular resistance, and thoracic fluid status. Under the previous coverage determination, effective July 1, 1999, use of TEB was covered for the “noninvasive diagnosis or monitoring of hemodynamics in patients with suspected or known cardiovascular disease.” In reconsidering this policy, CMS concluded that this use was neither sufficiently defined nor supported by available clinical literature to offer the guidance necessary for practitioners to determine when TEB would be covered for patient management. Therefore, CMS revised its coverage policy language in response to a request for reconsideration to offer more explicit guidance and clarity for coverage of TEB based on a complete and updated literature review.

Indications and Limitations of Coverage

A. Covered Indications

1. TEB is covered for the following uses:

- a. Differentiation of cardiogenic from pulmonary causes of acute dyspnea when medical history, physical examination, and standard assessment tools provide insufficient information, and the treating physician has determined that TEB hemodynamic data are necessary for appropriate management of the patient.
 - b. Optimization of atrioventricular (A/V) interval for patients with A/V sequential cardiac pacemakers when medical history, physical examination, and standard assessment tools provide insufficient information, and the treating physician has determined that TEB hemodynamic data are necessary for appropriate management of the patient.
 - c. Monitoring of continuous inotropic therapy for patients with terminal congestive heart failure, when those patients have chosen to die with comfort at home, or for patients waiting at home for a heart transplant.
 - d. Evaluation for rejection in patients with a heart transplant as a predetermined alternative to a myocardial biopsy. Medical necessity must be documented should a biopsy be performed after TEB.
 - e. Optimization of fluid management in patients with congestive heart failure when medical history, physical examination, and standard assessment tools provide insufficient information, and the treating physician has determined that TEB hemodynamic data are necessary for appropriate management of the patient.
2. Contractors have discretion to determine whether the use of TEB for the management of drug-resistant hypertension is reasonable and necessary. Drug resistant hypertension is defined as failure to achieve goal BP in patients who are adhering to full doses of an appropriate three-drug regimen that includes a diuretic.

B. Noncovered Indications

1. TEB is noncovered when used for patients:

- a. With proven or suspected disease involving severe regurgitation of the aorta;
- b. With minute ventilation (MV) sensor function pacemakers, since the device may adversely affect the functioning of that type of pacemaker;
- c. During cardiac bypass surgery; or
- d. In the management of all forms of hypertension (with the exception of drug-resistant hypertension as outlined above).

2. All other uses of TEB not otherwise specified remain non-covered.

(This NCD last reviewed January 2004.)

Transmittal Information

Transmittal Number

6

Coverage Transmittal Link

<https://www.cms.gov/Regulations-and-Guidance/Guidance/Transmittals/Downloads/R6NCD.pdf> 

Revision History

01/2004 - Offered more explicit guidance and clarification for coverage of TEB based on a complete and updated literature review. Effective date 01/23/2004. Implementation date 02/23/2004. (TN 6) (CR 2689)

04/1999 - Limited coverage for six uses. Contractors may cover additional uses when they believe there is sufficient evidence of medical effectiveness of such uses. Effective date 07/01/1999. (TN 109) (CR 827)

08/1989 - Excluded from coverage. Effective date 08/25/1989. (TN 38)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Electrical Bioimpedance for Cardiac Output Monitoring \(CAG-00001N\)](#) 
- [First reconsideration for Electrical Bioimpedance for Cardiac Output Monitoring \(CAG-00001R\)](#) 
- [Second reconsideration for Electrical Bioimpedance for Cardiac Output Monitoring \(CAG-00001R2\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)	3	11/24/2006 - N/A	View
Cardiac Output Monitoring by Thoracic Electrical Bioimpedance (TEB)	2	01/23/2004 - 11/24/2006	You are here
Cardiac Output Monitoring by Electrical Bioimpedance	1	07/01/1999 - 01/23/2004	View